Universal property of dependent pair types: the map from sections of a total space to pairs of dependent functions is an equivalence

The Univalent Foundations Program

hott-book:basics.tex:local-snapshot-20261014; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author theorem labelled thm:ttac (source lines 2418-2421); the proof span 2422-2436 is the author's proof environment, retained line for line.

Editorial note. Function extensionality is a background principle declared in the standard prerequisites.

Editorial note. Both retained contexts carry the citations used by the proof, so all references resolve inside the composed document.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the homotopical universal property of dependent pair types (the type-theoretic axiom of choice). Only the counted theorem and its author proof are retained; the author's axiom-of-choice commentary after the proof remains uncounted in the source.

Allowed standard prerequisites (editorial).

- Dependent pair types with their projections and dependent function types.
- The uniqueness principle for dependent pair types (the retained context).
- Function extensionality: pointwise identified dependent functions are identified.
- The uniqueness principle for functions applied to the pair of pointwise parts.

Jumping right to the dependently typed version, suppose we have a type X and type families $A : X \to \mathcal{U}$ and $P : \prod_{(x:X)} A(x) \to \mathcal{U}$. Then we have a function

$$\Big(\prod_{x:X} \sum_{(a:A(x))} P(x,a)\Big) \to \Big(\sum_{(g:\prod_{(x:X)} A(x))} \prod_{(x:X)} P(x, g(x))\Big). \tag{0.0.1}$$

Corollary 0.0.2. *For $z : \sum_{(x:A)} P(x)$, we have $z = (\mathsf{pr}_1(z), \mathsf{pr}_2(z))$.*

Theorem 0.0.3. (0.0.1) *is an equivalence.*

Proof. As before, we define a quasi-inverse to send (g, h) to the function $\lambda x.\, (g(x), h(x))$. Now given $f : \prod_{(x:X)} \sum_{(a:A(x))} P(x, a)$, the round-trip composite yields the function

$$\lambda x.\, (\mathsf{pr}_1(f(x)), \mathsf{pr}_2(f(x))). \tag{0.0.4}$$

Now for any $x : X$, by Corollary 0.0.2 (the uniqueness principle for Σ-types) we have

$$(\mathsf{pr}_1(f(x)), \mathsf{pr}_2(f(x))) = f(x).$$

Thus, by function extensionality, (0.0.4) is equal to f. On the other hand, given (g, h), the round-trip composite yields $(\lambda x.\, g(x), \lambda x.\, h(x))$, which is judgmentally equal to (g, h) as before. □